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Intergeniculate Leaflet Neurons: Structure, Connectivity, Markers, and Heterogeneity

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Introduction/Background

The intergeniculate leaflet (IGL) is a small thalamic structure in the circadian timing system that receives retinal input and is made up mostly of GABAergic neurons. Its network activity is strongly shaped by local inhibitory oscillations, with many neurons also expressing neuropeptides such as neuropeptide Y and enkephalins. (2 sources)

The intergeniculate leaflet is part of the brain system that links environmental light signals to circadian timing. In this role, it receives photic information from retinal ganglion cells, and this retinal drive is described as necessary for generating IGL network activity. At the same time, the leaflet is not just a passive relay: its local circuit is maintained by oscillatory activity of GABAergic interneurons, which make up most of the structure, and many of these neurons are associated with neuropeptide Y and enkephalins. Inputs from broader, non-specific brain projections can modulate this slow oscillatory circuit, although they do not appear to change the basic oscillatory pattern itself. (Lewandowski et al., 2004)

As a broad organizing principle, the IGL belongs to the largely GABAergic circadian timing network. Reports from rat indicate that many, and possibly all, IGL neurons contain GABA, placing inhibitory signaling at the center of leaflet function. This inhibitory architecture is also functionally important, because GABA_A receptor signaling has been implicated in the generation of ultradian rhythmic neuronal oscillations in the IGL. Together, these findings frame the IGL as a retinally driven, strongly inhibitory thalamic node that helps shape temporal and circadian processing rather than simply passing visual signals onward. (Blasiak et al., 2004) (Lewandowski et al., 2004)

Evidence

(Lewandowski et al., 2004)

Slow oscillation circuit of the intergeniculate leaflet.

> *"photic information flowing from ganglion cells of the retina is necessary for its generation. On the other hand, this circuit is maintained thanks to the oscillatory activity of GABAergic interneurons, the majority of which build up this structure, mainly in combination with neuropeptide Y and enkephalins. The activity of non-specific projections of the brain, whose terminals are present in the intergeniculate leaflet, modulates the slow oscillation circuit of the leaflet neurons, without changing its oscillatory pattern, though."*

(Blasiak et al., 2004)

Blockade of GABAA receptors disrupts isoperiodic neuronal oscillations in the intergeniculate leaflet of the rat.

"The majority of neurons in the circadian timing system are GABAergic. Many, if not all, neurons of the suprachiasmatic nucleus and intergeniculate leaflet contain GABA"

".These data suggest that GABA(A) receptors are involved in the generation of ultradian rhythmical neuronal oscillations in rat intergeniculate leaflet."

Anatomical location and structural organization of IGL neurons

The IGL is a thin, flat thalamic leaflet positioned between the dorsal and ventral divisions of the lateral geniculate nucleus and is often treated as a specialized retinorecipient zone related to the ventrolateral geniculate complex. Structurally, it is organized around two main GABAergic neuronal populations, ENK and NPY cells, which differ in projection targets and electrophysiological features. (6 sources)

The intergeniculate leaflet is a small bilateral thalamic structure in the lateral geniculate complex. In rodents, it is described as a narrow or flat leaflet interposed between the dorsal and ventral parts of the lateral geniculate nucleus, and it is retinorecipient, placing it anatomically at an interface between visual input pathways and circadian circuitry. Historically, it has often been considered a subdivision of the neighboring ventrolateral geniculate nucleus rather than a wholly separate nucleus. (Chrobok et al., 2016) (Palus et al., 2015) (Palus-Chramiec et al., 2019)

Comparative anatomy suggests that the precise boundaries of the IGL-like region can vary across species. In cat, neurons projecting to the SCN were found across the rostrocaudal extent of the medial ventral lateral geniculate nucleus, and this medial vLGN zone was proposed to be homologous to the rodent IGL. This supports the idea that the IGL is best understood as a specialized medial retinorecipient sector within the ventral geniculate region in some species. (Pu et al., 1996)

At the level of cellular organization, the IGL network is described as mainly GABAergic and dominated by two peptide-defined neuronal populations: enkephalinergic neurons and neuropeptide Y (NPY)-synthesizing neurons. These are not just molecular labels; they map onto different circuit roles and physiological signatures. ENK neurons are associated with the geniculo-geniculate projection linking the two IGLs, whereas NPY neurons are associated with the geniculohypothalamic projection to the SCN. (Chrobok et al., 2017) (Palus-Chramiec et al., 2019) (Palus et al., 2015) (Blasiak et al., 2013)

This structural division is also supported by intrinsic membrane properties. In rat, neurons that do not synthesize NPY were associated with a T-type calcium current, while all NPY-positive neurons examined showed a marked A-type potassium current. In later summaries of the same scheme, ENK neurons were linked to the T-type current and NPY neurons to the A-type current, reinforcing the idea that IGL organization includes both neurochemical and electrophysiological partitioning. (Chrobok et al., 2016) (Palus-Chramiec et al., 2019)

Beyond neurons, the IGL also contains a high density of GFAP-immunoreactive astrocytes. This suggests that its structural organization includes a prominent glial component in close association with its strongly inhibitory neuronal network. (Chrobok et al., 2017)

Evidence

(Chrobok et al., 2016)

Two distinct subpopulations of neurons in the thalamic intergeniculate leaflet identified by subthreshold currents.

"The intergeniculate leaflet (IGL) is a flat retinorecipient thalamic structure implicated in orchestrating circadian rhythm, historically considered to be a subdivision of the neighboring ventrolateral geniculate nucleus (VLG). IGL consists of two main neuronal subpopulations: enkephalinergic and neuropeptide Y (NPY)-synthesizing cells. These cell types have different functions, connectivity and firing pattern in vivo, which suggest that they have different membrane currents to support their functional differences"

".Our results suggest that IGL neurons can be divided into two subpopulations based on two ionic currents. A T-type calcium current (IT) was identified in neurons that do not synthesise NPY, whereas all NPY-positive neurons were found to express a marked A-type potassium current (IA)."

(Palus et al., 2015)

Electrophysiology of GABAergic transmission of single intergeniculate leaflet neurons in rat.

"The intergeniculate leaflet of the thalamus (IGL) is a narrow brain structure interlaid between the dorsal and ventral parts of the lateral geniculate nucleus of rodents (Moore et al., 1994). These small but important bilateral structures are known to modulate circadian rhythms, staying in reciprocal connection with the master biological clock generators -the suprachiasmatic nuclei of the hypothalamus (SCN) (Moore andCard 1994, Harrington 1997). Despite the presence of a few peptides located in IGL neurons, which are a key factor in differentiating the nucleus subpopulations, all IGL cells express the γ-aminobutyric acid (GABA) (Moore et al., 1993). The first neuronal group, distinguished by the expression of enkephalin (ENK), connects two contralateral IGLs (Takatsuji et al., 1989)Tohyama 1989, Harrington 1997), has a direct retinal innervation (Juhl et al., 2007)) and exhibit an infra-slow (<0.1 Hz) oscillatory pattern of neuronal activity in vivo (Błasiak et al., 2013). The other IGL subpopulation expresses an important brain modulator -neuropeptide Y (NPY) (Harrington, 1997). NPY-positive fibres originating from IGL are abundant in SCN (Card andMoore 1989, Saderi et al. 2013), and the NPY receptors coexist in pre-and postsynaptic sites on SCN neurons (Chen et al., 1996). It is known that NPY can produce circadian phase shifts (Jacob et al., 1999)(Lewandowski et al., 2002)(Lall et al., 2003). Putatively, the main function of IGL is collection and integration of the photic information from retina with non-photic cues mediated by non-specific brain systems (Harrington, 1997). It has been proposed that ENK neurons modulate the NPY cells activity and the comprehensive information is relayed to the SCN (Blasiak et al. 2006)."

(Palus-Chramiec et al., 2019)

Orexin A depolarises rat intergeniculate leaflet neurons through non-selective cation channels

"The intergeniculate leaflet (IGL) of the thalamus is a part of the circadian timing system that is strongly connected with the SCN, the master biological clock. This nucleus is located between the dorsal and ventral parts of the lateral geniculate nucleus. The neuronal network of the IGL mainly consists of the γ-aminobutyric acid (GABA)ergic neurons that co-expressed two distinct neuropeptides: enkephalin (ENK) or neuropeptide Y (NPY; (Moore et al., 1994). The characteristic feature of enkephalinergic neurons is the generation of infra-slow oscillations (ISO) in vivo. Moreover, this subpopulation of the IGL neurons is responsible for connecting IGLs in both hemispheres to each other (for review see: (Lewandowski et al., 2000). The second subpopulation of the IGL neurons (NPY-positive cells) connects the IGL with the SCN (Glass et al., 2010)Harrington, 1997;Moore & Card, 1994). Both IGL subpopulations are distinguished based on their electrophysiological properties. ENK neurons are characterised by the presence of the T-type calcium current (I T), whereas the NPY-positive cells exhibit the marked A-type"

(Pu et al., 1996)

Ventral lateral geniculate nucleus afferents to the suprachiasmatic nucleus in the cat.

"retrogradely labeled neurons were noted throughout the rostrocaudal extent of the medial vLGN following tracer injection into the SCN region. In addition, neuropeptide Y immunoreactive processes were also observed in the vLGN in this same medial zone and in the SCN. The data suggest that the medial zone of the cat vLGN may be homologous to the rodent intergeniculate leaflet (IGL)."

(Chrobok et al., 2017)

Disinhibition of the intergeniculate leaflet network in the WAG/Rij rat model of absence epilepsy

"The intergeniculate leaflet (IGL) of the thalamus is a retinorecipient structure implicated in orchestrating circadian rhythmicity. The IGL network is highly GABAergic and consists mainly of neuropeptide Y-synthesising and enkephalinergic neurons. A high density of GFAP-immunoreactive astrocytes has been observed in the IGL, with a probable function in guarding neuronal inhibition. Interestingly, putatively enkephalinergic IGL neurons generate action potentials with an infra-slow oscillatory (ISO) pattern in vivo in urethane anesthetised Wistar rats, under light-on conditions only."

(Blasiak et al., 2013)

Differential firing pattern and response to lighting conditions of rat intergeniculate leaflet neurons projecting to suprachiasmatic nucleus or contralateral intergeniculate leaflet.

"The intergeniculate leaflet (IGL) of the lateral geniculate body in the rat is a population of GABAergic neurons that can be divided into two, anatomically and neurochemically distinct populations. One population comprises neuropeptide-Y (NPY)-positive neurons that form the geniculohypothalamic tract innervating the suprachiasmatic nuclei (SCN) and the other population comprises enkephalin-positive (ENK) neurons giving rise to the geniculo-geniculate tract innervating the contralateral IGL (cIGL)"

".All IGL neurons identified as projecting to the contralateral IGL displayed infra-slow oscillatory activity (ISO; i.e. slow rhythmic bursts of action potentials). ISO of these neurons was sustained in the light and was diminished in the darkness. In contrast, all IGL neurons identified as projecting to the SCN displayed a low level of firing in the light and a majority of these cells increased firing in the darkness. All IGL neurons projecting to the SCN were characterised by an irregular pattern of firing in the light and dark. These data are the first to demonstrate that differentially projecting rat intergeniculate leaflet neurons are characterised by distinct firing patterns and opposite responses to light and dark conditions."

Core neuronal classes, neurotransmitters, neuropeptides, and physiological signatures

IGL neurons are predominantly GABAergic and are usually discussed as two main peptide-defined classes: NPY neurons and enkephalinergic neurons. These classes differ not only in peptide content and outputs, but also in firing mode, responses to light/dark, and characteristic membrane currents. (8 sources)

At the transmitter level, the IGL is built mainly from GABAergic neurons, and many reports state that many or even most IGL neurons contain GABA. GABA is also functionally important, because GABA_A receptor signaling has been linked to the generation of rhythmic oscillatory activity in the IGL rather than serving only a conventional inhibitory role. (Blasiak et al., 2004) (Blasiak et al._1, 2013)

Within this GABAergic network, the best established neuronal classes are the NPY-positive cells and the enkephalin-positive cells. These are repeatedly described as the two main IGL subpopulations, and in both cases the peptide signal is typically discussed as co-existing with GABA rather than replacing it. (Chrobok et al., 2016) (Palus et al., 2017) (Chrobok et al., 2017) (Palus-Chramiec et al., 2019)

These classes also differ in intrinsic physiology. NPY-synthesizing neurons are associated with a prominent A-type potassium current (I_A), while non-NPY neurons, later summarized as the enkephalinergic class, show a T-type calcium current (I_T). This provides a membrane basis for their different activity patterns in vivo. (Chrobok et al., 2016) (Palus-Chramiec et al., 2019)

Their firing signatures are strongly distinct. Neurons projecting to the contralateral IGL, corresponding to the ENK/geniculo-geniculate class, show infra-slow oscillatory activity with slow rhythmic bursting; this activity persists in light and is reduced in darkness. By contrast, neurons projecting to the SCN, corresponding to the NPY/geniculohypothalamic class, show relatively low and irregular firing in the light, and many increase firing in darkness. (Blasiak et al., 2013) (Chrobok et al., 2017) (Palus-Chramiec et al., 2019)

The enkephalin system appears to have both neuromodulatory and local inhibitory roles inside the IGL. Met-enkephalin robustly hyperpolarized both putative NPY and putative ENK neurons, and it suppressed both GABAergic and glutamatergic synaptic transmission; in that study, the \mu-opioid receptor was the functionally predominant opioid receptor. These data support the idea that enkephalinergic neurons can act as strong inhibitory interneuron-like elements within and across the two IGLs. (Palus et al., 2017)

Although the NPY/ENK division is the core framework, physiological recordings suggest additional heterogeneity. In the IGL/VLG region, three classes were described based on NPY immunolabeling and light responses: Type I neurons lacked NPY and showed sustained activation during retinal illumination; Type II neurons contained strong somatic NPY and showed mixed light responses, including suppression or complex responses; and Type III neurons showed patchy surface and intracellular NPY labeling, were activated by light like Type I cells, but were nearly silent in the intervening dark period, consistent with possible suppression by NPY input from other IGL neurons. (Thankachan et al., 2005)

A further sign that the classical two-class model is incomplete is the report of a third IGL cell group in rat with unknown neurotransmitter content. In functional tests, identified NPY-positive neurons were excited by serotonergic stimulation (R3/I5), whereas neurons of unidentified transmitter content could either depolarize or reduce firing and hyperpolarize, again pointing to physiological diversity beyond the NPY and ENK groups. (Blasiak et al._1, 2013)

Evidence

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Blockade of GABAA receptors disrupts isoperiodic neuronal oscillations in the intergeniculate leaflet of the rat.

> *"The majority of neurons in the circadian timing system are GABAergic. Many, if not all, neurons of the suprachiasmatic nucleus and intergeniculate leaflet contain GABA"*

".These data suggest that GABA(A) receptors are involved in the generation of ultradian rhythmical neuronal oscillations in rat intergeniculate leaflet."

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Relaxin-3 innervation of the intergeniculate leaflet of the rat thalamus – neuronal tract-tracing and in vitro electrophysiological studies

"The intergeniculate leaflet (IGL) of the lateral geniculate body of rodents, a homologue of the pre-geniculate nucleus in primates (Moore, 1989), is a structure known for its role in the regulation of circadian rhythmicity (Morin, 1994))"

".The most abundant neurotransmitter synthesized by IGL neurons is c-aminobutyric acid (Moore & Speh, 1993). In the majority of IGL neurons, c-aminobutyric acid is co-localized with neuropeptide Y (NPY) or enkephalin. In the rat IGL, a third cell group is present with unknown neurotransmitter content (Moore & Card, 1994). NPY-containing IGL neurons form the geniculo-hypothalamic tract that terminates in the suprachiasmatic nucleus (SCN), considered to be the main circadian 'pacemaker' in the circadian timing system"

".Identified, neuropeptide Y-positive IGL neurons, known to influence suprachiasmatic nucleus activity, were excited by R3/I5, whereas neurons of unidentified neurotransmitter content were either depolarized or displayed a decrease in action potential firing and/or membrane potential hyperpolarization."

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Two distinct subpopulations of neurons in the thalamic intergeniculate leaflet identified by subthreshold currents.

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Enkephalin and neuropeptide-Y interaction in the intergeniculate leaflet network, a part of the mammalian biological clock.

"The intergeniculate leaflet (IGL) is a flat thalamic nucleus implicated in the modulation of circadian rhythmicity. In rat, two main GABAergic subpopulations can be distinguished in the IGL: neurons synthesizing neuropeptide Y (NPY), which directly innervates the suprachiasmatic nuclei, and enkephalinergic cells, which connect contralaterally located leaflets"

".Moreover, we investigated the type of opioid receptor activated by enkephalin and showed that the μ-receptor is functionally predominant in the IGL. The application of met-enkephalin not only robustly hyperpolarized IGL neurons (both putatively NPY-synthesizing and putatively enkephalinergic neurons), but it also was able to inhibit GABAergic and glutamatergic synaptic transmission. Based on this and previous studies, we hypothesize that IGL enkephalinergic neurons may act as powerful interneurons that inhibit themselves and NPY-synthesizing neurons, also in the contralaterally located IGL."

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(Thankachan et al., 2005)

Juxtacellular Recording/Labeling Analysis of Physiological and Anatomical Characteristics of Rat Intergeniculate Leaflet Neurons

"The thalamic intergeniculate leaflet (IGL) is involved in mediating effects of both photic and nonphotic stimuli on mammalian circadian rhythms. IGL neurons containing neuropeptide Y (NPY) have been implicated in mediating nonphotic effects, but little is known about those involved in photic entrainment"

". Three classes of neurons were identified in the IGL/VLG. Type I neurons lacked NPY and showed sustained activations during retinal illumination and moderate firing rates in darkness. Type II neurons contained large amounts of NPY throughout the soma and showed varied responses to illumination: suppression, complex responses, or no response. Type III neurons had patches of NPY both on the external soma surface and within the soma, apparently representing internalization of NPY. Type III neurons resembled type I cells in their sustained activation by illumination but were virtually silent during the intervening dark period. These neurons appear to receive NPY input, presumably from other IGL cells, which may suppress their activity during darkness. These results demonstrate the presence of several classes of neurons in the IGL defined by their functional and anatomical features and reinforce the role of the IGL/VLG complex in integrating photic and nonphotic inputs to the circadian system."

Connectivity and circuit functions of IGL neuronal subtypes

IGL connectivity is built around two main GABAergic projection systems: NPY neurons that send output to the SCN, and ENK neurons that link the two IGLs across hemispheres. These two pathways appear to serve different circuit roles, with ENK cells helping organize local and bilateral inhibitory network rhythms, while NPY cells convey integrated photic and non-photic timing signals to the SCN. (9 sources)

A central wiring scheme of the rat IGL is the division between NPY-positive neurons and enkephalin-positive neurons. NPY neurons form the geniculohypothalamic tract and innervate the SCN, whereas ENK neurons form the geniculo-geniculate projection to the contralateral IGL. This connectional split is one of the clearest subtype-level distinctions in the nucleus and aligns with the different physiological profiles described above. (Blasiak et al., 2013) (Palus-Chramiec et al., 2019) (Blasiak et al._1, 2013)

Functionally, the NPY SCNpathway places the IGL in a direct position to modulate the master circadian clock. Reviews and experimental work describe the IGL as strongly connected with the SCN and involved in both photic and non-photic control of circadian rhythms. NPY fibers from the IGL are abundant in the SCN, NPY receptors are present at both pre- and postsynaptic sites in SCN neurons, and NPY signaling from this pathway can shift circadian phase. This supports the view that IGL output is not simply sensory relay, but a modulatory input that can alter SCN state and timing. (Palus et al., 2015) (Chen et al., 1996)

The ENK system appears to support a different circuit role. ENK neurons connect the left and right IGLs, receive direct retinal innervation, and are the subtype most closely tied to infra-slow oscillatory population activity. In vivo, all identified neurons projecting to the contralateral IGL showed infra-slow rhythmic bursting that persisted in light and was reduced in darkness, suggesting that this bilateral ENK-linked pathway may help sustain or coordinate rhythmic inhibitory activity across the two leaflets. (Blasiak et al., 2013) (Palus et al., 2015) (Palus-Chramiec et al., 2019)

By contrast, identified SCN-projecting IGL neurons had low and irregular firing in light, and most increased firing in darkness. This opposite light/dark response relative to contralateral IGL-projecting cells suggests that the two pathways carry different temporal information: the ENK/bilateral system is linked to ongoing rhythmic network state, while the NPY/SCN system may convey a more state-dependent output to the hypothalamic clock. (Blasiak et al., 2013)

Several studies frame the IGL as an integrator of retinal photic input with broader non-photic brain signals. Retinal ganglion cell input is described as necessary for generating IGL network activity, while non-specific brain projections modulate the slow oscillation circuit without changing its basic oscillatory pattern. In this model, the IGL uses local GABAergic circuitry to combine photic drive with non-photic influences and then routes that integrated signal onward, especially to the SCN through NPY neurons. (Lewandowski et al., 2004) (Palus et al., 2015)

There is also evidence for local and bilateral inhibitory control within the IGL itself. Met-enkephalin hyperpolarized both putative NPY and putative ENK neurons and inhibited both GABAergic and glutamatergic synaptic transmission, with the μ-opioid receptor being the predominant functional opioid receptor in the IGL. These data led to the proposal that ENK neurons can act as powerful interneuron-like elements that inhibit themselves, inhibit NPY neurons, and exert similar control in the contralateral IGL through the geniculo-geniculate projection. This gives a concrete circuit mechanism by which the ENK population could gate IGL output to the SCN. (Palus et al., 2017)

The IGL/VLG region also shows subtype-specific response patterns that imply more complex internal connectivity than the simple ENK-versus-NPY scheme. In one study, Type I neurons lacked NPY and were steadily activated by retinal illumination; Type II neurons contained strong somatic NPY and showed mixed light responses; and Type III neurons showed patchy surface and intracellular NPY labeling, were activated by light, but were nearly silent in darkness. The authors suggested that Type III cells may receive NPY input from other IGL neurons that suppresses their dark-period activity, providing direct evidence for peptide-defined recurrent interactions inside the IGL/VLG complex. (Thankachan et al., 2005)

Additional inputs from outside the classic retinal pathway likely add further state dependence to IGL output. For example, identified NPY-positive IGL neurons were excited by stimulation of the serotonergic dorsal and median raphe region (R3/I5), while a third group of IGL neurons with unknown transmitter content showed variable responses, including depolarization or reduced firing with hyperpolarization. This suggests that neuromodulatory brainstem input can recruit specific IGL output channels differently, again consistent with a role in integrating non-photic signals before they influence circadian targets. (Blasiak et al._1, 2013)

Across species, the SCN-projecting part of this circuit may extend into the medial ventral lateral geniculate region rather than always appearing as a sharply separate leaflet. In cat, SCN-projecting neurons and NPY-immunoreactive processes were found throughout the medial vLGN, and this zone was proposed to be homologous to the rodent IGL. This suggests that the core connectional motif—retinorecipient geniculate neurons, NPY signaling, and SCN projection—is conserved even if the precise anatomical boundaries vary. (Pu et al., 1996)

Evidence

(Blasiak et al., 2013)

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(Palus et al., 2015)

Electrophysiology of GABAergic transmission of single intergeniculate leaflet neurons in rat.

"The intergeniculate leaflet of the thalamus (IGL) is a narrow brain structure interlaid between the dorsal and ventral parts of the lateral geniculate nucleus of rodents (Moore et al., 1994). These small but important bilateral structures are known to modulate circadian rhythms, staying in reciprocal connection with the master biological clock generators -the suprachiasmatic nuclei of the hypothalamus (SCN) (Moore andCard 1994, Harrington 1997). Despite the presence of a few peptides located in IGL neurons, which are a key factor in differentiating the nucleus subpopulations, all IGL cells express the γ-aminobutyric acid (GABA) (Moore et al., 1993). The first neuronal group, distinguished by the expression of enkephalin (ENK), connects two contralateral IGLs (Takatsuji et al., 1989)Tohyama 1989, Harrington 1997), has a direct retinal innervation (Juhl et al., 2007)) and exhibit an infra-slow (<0.1 Hz) oscillatory pattern of neuronal activity in vivo (Błasiak et al., 2013). The other IGL subpopulation expresses an important brain modulator -neuropeptide Y (NPY) (Harrington, 1997). NPY-positive fibres originating from IGL are abundant in SCN (Card andMoore 1989, Saderi et al. 2013), and the NPY receptors coexist in pre-and postsynaptic sites on SCN neurons (Chen et al., 1996). It is known that NPY can produce circadian phase shifts (Jacob et al., 1999)(Lewandowski et al., 2002)(Lall et al., 2003). Putatively, the main function of IGL is collection and integration of the photic information from retina with non-photic cues mediated by non-specific brain systems (Harrington, 1997). It has been proposed that ENK neurons modulate the NPY cells activity and the comprehensive information is relayed to the SCN (Blasiak et al. 2006)."

(Chen et al., 1996)

Multiple NPY Receptors Coexist in Pre- and Postsynaptic Sites: Inhibition of GABA Release in Isolated Self-Innervating SCN Neurons

"Although NPY has been shown to influence the action of many transmitters in the brain, modulation of GABA, the primary inhibitory transmitter, has not been detected with electrophysiology. Using whole-cell patch-clamp recording, we found that NPY has a large modulatory effect on GABAergic neurons of the suprachiasmatic nucleus (SCN) that act as the circadian clock in the mammalian brain. NPY, acting at both Y1- and Y2-like receptors, reduced the frequency of spontaneous miniature inhibitory postsynaptic currents while having little effect on the postsynaptic GABA receptors, suggesting a presynaptic mechanism of NPY action. In single self-innervating neurons, application of either Y1 or Y2 agonists to the same neuron significantly inhibited the evoked autaptic GABA

release. The use of single-neuron microcultures has allowed the demonstration that a single peptide, NPY, has two different receptors coded for by different genes in the same axon terminal. The Y1 and Y2 agonists also inhibited whole-cell calcium currents when applied to the same neuron, indicating a coexistence of Y1- and Y2-like receptors in the postsynaptic cell body. The self-innervating cell model we use here may be applicable generally for discriminating presynaptic versus postsynaptic actions of other neurotransmitters and neuromodulators and locating their subtype receptors."

(Lewandowski et al., 2004)

Slow oscillation circuit of the intergeniculate leaflet.

"photic information flowing from ganglion cells of the retina is necessary for its generation. On the other hand, this circuit is maintained thanks to the oscillatory activity of GABAergic interneurons, the majority of which build up this structure, mainly in combination with neuropeptide Y and enkephalins. The activity of non-specific projections of the brain, whose terminals are present in the intergeniculate leaflet, modulates the slow oscillation circuit of the leaflet neurons, without changing its oscillatory pattern, though."

(Palus et al., 2017)

Enkephalin and neuropeptide-Y interaction in the intergeniculate leaflet network, a part of the mammalian biological clock.

"The intergeniculate leaflet (IGL) is a flat thalamic nucleus implicated in the modulation of circadian rhythmicity. In rat, two main GABAergic subpopulations can be distinguished in the IGL: neurons synthesizing neuropeptide Y (NPY), which directly innervates the suprachiasmatic nuclei, and enkephalinergic cells, which connect contralaterally located leaflets"

".Moreover, we investigated the type of opioid receptor activated by enkephalin and showed that the μ-receptor is functionally predominant in the IGL. The application of met-enkephalin not only robustly hyperpolarized IGL neurons (both putatively NPY-synthesizing and putatively enkephalinergic neurons), but it also was able to inhibit GABAergic and glutamatergic synaptic transmission. Based on this and previous studies, we hypothesize that IGL enkephalinergic neurons may act as powerful interneurons that inhibit themselves and NPY-synthesizing neurons, also in the contralaterally located IGL."

(Thankachan et al., 2005)

Juxtacellular Recording/Labeling Analysis of Physiological and Anatomical Characteristics of Rat Intergeniculate Leaflet Neurons

"The thalamic intergeniculate leaflet (IGL) is involved in mediating effects of both photic and nonphotic stimuli on mammalian circadian rhythms. IGL neurons containing neuropeptide Y (NPY) have been implicated in mediating nonphotic effects, but little is known about those involved in photic entrainment"

". Three classes of neurons were identified in the IGL/VLG. Type I neurons lacked NPY and showed sustained activations during retinal illumination and moderate firing rates in darkness. Type II neurons contained large amounts of NPY throughout the soma and showed varied responses to illumination: suppression, complex responses, or no response. Type III neurons had patches of NPY both on the external soma surface and within the soma, apparently representing internalization of NPY. Type III neurons resembled type I cells in their sustained activation by illumination but were virtually silent during the intervening dark period. These neurons appear to receive NPY input, presumably from other IGL cells, which may suppress their activity during darkness. These results demonstrate the presence of several classes of neurons in the IGL defined by their functional and anatomical features and reinforce the role of the IGL/VLG complex in integrating photic and nonphotic inputs to the circadian system."

(Pu et al., 1996)

Ventral lateral geniculate nucleus afferents to the suprachiasmatic nucleus in the cat.

"retrogradely labeled neurons were noted throughout the rostrocaudal extent of the medial vLGN following tracer injection into the SCN region. In addition, neuropeptide Y immunoreactive processes were also observed in the vLGN in this same medial zone and in the SCN. The data suggest that the medial zone of the cat vLGN may be homologous to the rodent intergeniculate leaflet (IGL)."

Markers and their anatomical context

The most useful markers for IGL neurons in the cited literature are GABA, NPY, and enkephalin, with GFAP marking the dense astrocyte population. In this region, marker meaning depends strongly on anatomical context: some labels are used for the IGL proper, while others come from the broader IGL/vLGN complex or from vLGN sublaminae that may overlap with IGL-homologous territory. (7 sources)

- **GABA**
- **Marker type:** protein/immunochemical marker of transmitter phenotype.
- **What it marks in context:** a broad inhibitory identity for most or possibly nearly all IGL neurons.

- **Anatomical context:** IGL as part of the circadian timing system.
- **Usefulness/limits:** this is a population-level marker, not a subtype-specific one. It tells you that the neuron is part of the dominant inhibitory network, but not whether it belongs to the NPY, ENK, or other reported classes. (Blasiak et al., 2004) (Blasiak et al._1, 2013)
- **NPY (neuropeptide Y)**
- **Marker type:** peptide/protein expression; in the cited work this is used mainly by immunolabeling or identification of NPY-positive neurons.
- **What it marks in context:** a major IGL neuronal class, usually GABAergic, associated with the geniculohypothalamic projection to the SCN.
- **Anatomical context:** IGL proper, and also the broader IGL/VLG complex where NPY labeling helped separate functional response classes.
- **Functional marker context:** in rat IGL, NPY-positive neurons are the class linked to SCN output and can be identified physiologically as a group distinct from ENK neurons. (Blasiak et al._1, 2013) (Palus-Chramiec et al., 2019)
- **Subtype/anatomical nuance:** in the IGL/VLG study, strong somatic NPY labeling defined **Type II** neurons, whereas **Type III** neurons showed patchy NPY labeling on the soma surface and within the soma, interpreted as possible peptide uptake/internalization rather than simple cell-intrinsic peptide content. That means NPY labeling can mark both an NPY-producing class and cells that appear to receive NPY input, depending on staining pattern. (Thankachan et al., 2005)
- **Enkephalin (ENK)**
- **Marker type:** peptide/protein expression.
- **What it marks in context:** the second major IGL neuronal class, again usually discussed within the GABAergic population.
- **Anatomical context:** IGL proper.
- **Functional marker context:** ENK neurons are the class linked to the bilateral geniculo-geniculate pathway and to infra-slow oscillatory firing. In practice, ENK is used as a subtype marker together with physiology and projection target. (Chrobok et al., 2017) (Palus-Chramiec et al., 2019)
- **Usefulness/limits:** ENK is a meaningful subtype marker in the IGL, but the cited material does not give a more refined molecular subdivision within the ENK class.
- **GFAP**
- **Marker type:** protein/immunoreactive astrocyte marker.
- **What it marks in context:** astrocytes rather than neurons.
- **Anatomical context:** IGL proper.
- **Functional marker context:** a high density of GFAP-immunoreactive astrocytes has been reported in IGL, suggesting a prominent glial component around the inhibitory network. This is important for tissue organization, but GFAP is not a neuronal subtype marker. (Chrobok et al., 2017)
- **Electrophysiological features used alongside markers**
- **Marker type:** not molecular markers, but repeatedly used as subtype-defining features.
- **What they distinguish:** ENK neurons are associated with **T-type calcium current (I_T)**, while NPY-positive

neurons show a marked **A-type potassium current**.

- **Anatomical context:** rat IGL.
- **Why note this here:** in this literature, peptide markers and intrinsic physiology are tightly paired, so subtype identification often uses both. (Palus-Chramiec et al., 2019)
- **"Unknown neurotransmitter content" group**
- **Marker type:** effectively a negative category, not a positive marker.
- **What it marks in context:** a third reported IGL cell group that is not assigned to the NPY or ENK classes in the cited rat work.
- **Anatomical context:** rat IGL.
- **Usefulness/limits:** this is evidence for heterogeneity, but not yet a usable molecular marker set. (Blasiak et al._1, 2013)
- **NPY-pattern-based classes in the IGL/VLG complex**
- **Marker type:** protein/immunolabel pattern plus physiology.
- **What they mark in context:**
- **Type I:** NPY-negative; sustained activation during retinal illumination.
- **Type II:** strong somatic NPY; varied light responses.
- **Type III:** patchy surface and intracellular NPY; sustained light activation but near silence in dark.
- **Anatomical context:** broader **IGL/VLG** region, not strictly isolated IGL.
- **Why this matters:** this is one of the clearest examples where the same molecular marker, NPY, functions differently depending on staining pattern and local circuitry. It also shows that "NPY-positive" is not a single uniform physiological category. (Thankachan et al., 2005)
- **Transcript-based and transgene-based markers from ventral LGN studies**
- **Marker type:** transcript expression and, in many cases, transgene/reporter-assisted labeling in mouse vLGN studies.
- **What they mark in context:** distinct GABAergic cell types and sublaminae within external and internal vLGN, including four transcriptomically distinct retinorecipient sublaminae in vLGNe.
- **Anatomical context: vLGN**, especially vLGNe, rather than definitively delimited IGL.
- **Why include with caution:** the IGL has often been treated as related to or partly continuous with medial vLGN territory, so these data are relevant to the broader anatomical neighborhood. But the cited study does **not** specifically assign these transcriptomic marker-defined sublaminae to the IGL proper. Thus they should be treated as candidate context for IGL-adjacent or IGL-homologous territories, not as established IGL neuron markers. (Sabbagh et al., 2020) (Gouwens et al., 2020)
- **Summary of best-supported markers for IGL neurons**
- For **neurons overall**, the most reliable marker framework in the provided literature is **GABA + peptide class**, with the two main peptide markers being **NPY** and **ENK**. (Blasiak et al., 2004) (Chrobok et al., 2017) (Blasiak et al._1, 2013) (Palus-Chramiec et al., 2019)
- For **glia**, **GFAP** marks the dense astrocyte population in IGL. (Chrobok et al., 2017)

- For **finer molecular subtyping**, the strongest transcriptomic evidence in the provided references is from the neighboring **vLGN**, not from a clean single-cell dataset dedicated to IGL neurons themselves. (Sabbagh et al., 2020)

Evidence

(Blasiak et al., 2004)

Blockade of GABAA receptors disrupts isoperiodic neuronal oscillations in the intergeniculate leaflet of the rat.

> *"The majority of neurons in the circadian timing system are GABAergic. Many, if not all, neurons of the suprachiasmatic nucleus and intergeniculate leaflet contain GABA"*
>
> *".These data suggest that GABA(A) receptors are involved in the generation of ultradian rhythmical neuronal oscillations in rat intergeniculate leaflet."*

(Blasiak et al._1, 2013)

Relaxin-3 innervation of the intergeniculate leaflet of the rat thalamus – neuronal tract-tracing and in vitro electrophysiological studies

> *"The intergeniculate leaflet (IGL) of the lateral geniculate body of rodents, a homologue of the pre-geniculate nucleus in primates (Moore, 1989), is a structure known for its role in the regulation of circadian rhythmicity (Morin, 1994))"*
>
> *".The most abundant neurotransmitter synthesized by IGL neurons is c-aminobutyric acid (Moore & Speh, 1993). In the majority of IGL neurons, c-aminobutyric acid is co-localized with neuropeptide Y (NPY) or enkephalin. In the rat IGL, a third cell group is present with unknown neurotransmitter content (Moore & Card, 1994). NPY-containing IGL neurons form the geniculo-hypothalamic tract that terminates in the suprachiasmatic nucleus (SCN), considered to be the main circadian 'pacemaker' in the circadian timing system"*
>
> *".Identified, neuropeptide Y-positive IGL neurons, known to influence suprachiasmatic nucleus activity, were excited by R3/I5, whereas neurons of unidentified neurotransmitter content were either depolarized or displayed a decrease in action potential firing and/or membrane potential hyperpolarization."*

(Palus-Chramiec et al., 2019)

Orexin A depolarises rat intergeniculate leaflet neurons through non-selective cation channels

> *"The intergeniculate leaflet (IGL) of the thalamus is a part of the circadian timing system that is strongly connected with the SCN, the master biological clock. This nucleus is located between the dorsal and ventral parts of the lateral geniculate nucleus. The neuronal network of the IGL mainly consists of the γ-aminobutyric acid (GABA)ergic neurons that co-expressed two distinct neuropeptides: enkephalin (ENK) or neuropeptide Y (NPY; (Moore et al., 1994). The characteristic feature of enkephalinergic neurons is the generation of infra-slow oscillations (ISO) in vivo. Moreover, this subpopulation of the IGL neurons is responsible for connecting IGLs in both hemispheres to each other (for review see: (Lewandowski et al., 2000). The second subpopulation of the IGL neurons (NPY-positive cells) connects the IGL with the SCN (Glass et al., 2010)Harrington, 1997;Moore & Card, 1994). Both IGL subpopulations are distinguished based on their electrophysiological properties. ENK neurons are characterised by the presence of the T-type calcium current (I T), whereas the NPY-positive cells exhibit the marked A-type"*

(Thankachan et al., 2005)

Juxtacellular Recording/Labeling Analysis of Physiological and Anatomical Characteristics of Rat Intergeniculate Leaflet Neurons

> *"The thalamic intergeniculate leaflet (IGL) is involved in mediating effects of both photic and nonphotic stimuli on mammalian circadian rhythms. IGL neurons containing neuropeptide Y (NPY) have been implicated in mediating nonphotic effects, but little is known about those involved in photic entrainment"*
>
> *". Three classes of neurons were identified in the IGL/VLG. Type I neurons lacked NPY and showed sustained activations during retinal illumination and moderate firing rates in darkness. Type II neurons contained large amounts of NPY throughout the soma and showed varied responses to illumination: suppression, complex responses, or no response. Type III neurons had patches of NPY both on the external soma surface and within the soma, apparently representing internalization of NPY. Type III neurons resembled type I cells in their sustained activation by illumination but were virtually silent during the intervening dark period. These neurons appear to receive NPY input, presumably from other IGL cells, which may suppress their activity during darkness. These results demonstrate the presence of several classes of neurons in the IGL defined by their functional and anatomical features and reinforce the role of the IGL/VLG complex in integrating photic and nonphotic inputs to the circadian system."*

(Chrobok et al., 2017)

Disinhibition of the intergeniculate leaflet network in the WAG/Rij rat model of absence epilepsy

> *"The intergeniculate leaflet (IGL) of the thalamus is a retinorecipient structure implicated in orchestrating circadian rhythmicity. The IGL network is highly GABAergic and consists mainly of neuropeptide Y-synthesising and enkephalinergic neurons. A high density of GFAP-immunoreactive astrocytes has been observed in the IGL, with a probable function in guarding neuronal inhibition. Interestingly, putatively enkephalinergic IGL neurons generate action potentials with an infra-slow oscillatory (ISO) pattern in vivo in urethane anesthetised Wistar rats, under light-on conditions only."*

(Sabbagh et al., 2020)

Diverse GABAergic neurons organize into subtype-specific sublaminae in the ventral lateral geniculate nucleus

> *"we labeled neurons in vLGN with canonical and novel cell type-specific markers and studied their spatial distribution and morphoelectric properties. Not only did we find a high percentage of cells in vLGN to be GABAergic, we discovered transcriptomically distinct GABAergic cell types reside in the two major laminae of vLGN, the retinorecipient, external vLGN (vLGNe) and the non-retinorecipient, internal vLGN (vLGNi). Within vLGNe, we identified transcriptionally distinct subtypes of GABAergic cells that are distributed into four adjacent sublaminae. Using trans-synaptic viral tracing and in vitro electrophysiology, we found cells in each these vLGNe sublaminae receive monosynaptic inputs from the retina"*
>
> *". In this study, we describe subtypes of transcriptomically distinct GABAergic neurons that populate the vLGN and organize into discrete, adjacent sublaminae in the vLGNe. Taken together, our results show four subtype-specific sublaminae of retinorecipient neurons in vLGNe"*
>
> *". Here, we used spatial distribution and transcriptional profiles to classify neurons into distinct subtypes. Molecular markers remain at present a leading characteristic for such classifications, though they are not without their limitations. It may well be that vLGN neurons can be further subdivided if classified by 2-3 molecular markers rather than one (as we observed in Spp1 + and Pvalb + neurons). However, the very fact that differential expression of one molecule was sufficient to differentiate two vLGN populations is a strong indicator that the nucleus as a whole is more diversely populated than previously appreciated"*
>
> *". A set of recent studies represent a major step towards this goal by utilizing Patch-seq to simultaneously characterize cortical GABAergic neurons electrophysiologically, morphologically, and transcriptomically (Gouwens et al., 2020)Scala et al., 2020)."*

(Gouwens et al., 2020)

Toward an integrated classification of neuronal cell types: morphoelectric and transcriptomic characterization of individual GABAergic cortical neurons

> *"Neurons are frequently classified into distinct groups or cell types on the basis of structural, physiological, or genetic attributes. To better constrain the definition of neuronal cell types, we characterized the transcriptomes and intrinsic physiological properties of over 3,700 GABAergic mouse visual cortical neurons and reconstructed the local morphologies of 350 of those neurons. We found that most transcriptomic types (t-types) occupy specific laminar positions within mouse visual cortex, and many of those t-types exhibit consistent electrophysiological and morphological features. We observed that these properties could vary continuously between t-types, which limited the ability to predict specific t-types from other data modalities. Despite that, the data support the presence of at least 20 interneuron met-types that have congruent morphological, electrophysiological, and transcriptomic properties. Highlights Patch-seq data obtained from >3,700 GABAergic cortical interneurons Comprehensive characterization of morpho-electric features of transcriptomic types 20 interneuron met-types that have congruent properties across data modalities Different Sst met-types preferentially innervate different cortical layers"*

Heterogeneity and subtype evidence, including transcriptomic data

The best-supported model of IGL neuronal heterogeneity is still a small set of peptide- and projection-defined GABAergic classes rather than a large transcriptomic taxonomy. Strongest evidence supports at least two main classes, NPY and ENK, with additional functional subtypes in the broader IGL/VLG region and transcriptomic diversification shown clearly in adjacent vLGN but not yet resolved specifically for IGL proper. (5 sources)

Taken together, the cited work supports a layered view of IGL heterogeneity. At the core is a robust two-class scheme in which IGL neurons split into NPY-synthesizing cells and enkephalinergic cells, and this division is supported independently by neurochemistry, projection target, firing pattern, and intrinsic membrane currents. In rat, NPY-positive neurons project to the SCN, while ENK-positive neurons project to the contralateral IGL; these two groups also differ in light/dark firing and in the ionic conductances associated with them, with NPY cells linked to a marked A-type potassium current and non-NPY/ENK cells linked to a T-type calcium current. This is the clearest subtype framework available for the IGL proper. (Chrobok et al., 2016) (Blasiak et al., 2013)

At the same time, the two-class model is not the whole story. In the broader IGL/VLG complex, at least three physiologically and anatomically distinguishable neuron classes were defined using NPY immunolabeling pattern plus retinal light response. Type I cells were NPY-negative and light-activated; Type II cells had strong somatic NPY and showed mixed responses to illumination; Type III cells showed patchy surface and intracellular NPY labeling, were also light-activated, but were nearly silent in darkness, consistent with receipt of NPY input from other IGL neurons. This is important because it shows that even within "NPY-related" cells there may be more than one biological state: an NPY-producing class and a class marked by NPY-associated uptake or receptor-driven peptide internalization rather than simple intrinsic peptide expression. (Thankachan et al., 2005)

So, for IGL proper, the current evidence fits best with **at least two major classes plus additional unresolved heterogeneity**. The unresolved part could reflect cells not cleanly captured by NPY versus ENK labels, or finer subdivisions within those peptide classes. The older physiological and anatomical data already point in that direction, because projection-defined classes differ not only in output target but also in ongoing activity regime, suggesting that subtype identity in IGL is tied to circuit role as much as to peptide content. (Blasiak et al., 2013) (Chrobok et al., 2016)

For transcriptomics, the key limitation is that the provided references do **not** include a single-cell or spatial transcriptomic atlas dedicated specifically to the IGL proper. The closest relevant dataset is from mouse vLGN, where transcriptomically distinct GABAergic neuron types were found in both external and internal laminae, and four adjacent retinorecipient sublaminae were identified within vLGNe. Those vLGN cell types were defined using transcript expression, spatial arrangement, and reporter-assisted marker analysis, and were linked to distinct morphoelectric features; all four retinorecipient sublaminae received monosynaptic retinal input. This is highly relevant for the IGL because the IGL has often been treated as closely related to, or historically part of, ventral geniculate territory. But the evidence should be used carefully: these transcriptomic sublaminae were established for **vLGN**, not assigned directly to a delimited IGL neuron taxonomy. (Sabbagh et al., 2020)

In marker terms, that means the heterogeneity evidence is uneven across anatomical scales. In IGL itself, the strongest subtype markers remain **protein-level peptide markers** such as NPY and ENK, interpreted in the context of projection target and physiology. In neighboring vLGN, **transcript-based markers** reveal a richer GABAergic cell-type landscape and suggest that retinorecipient geniculate territory is more diverse than the classic IGL two-class scheme implies. But based on the provided papers, one cannot yet map those vLGN transcriptomic cell types one-to-one onto ENK, NPY, Type I, Type II, or Type III IGL classes with confidence. (Sabbagh et al., 2020) (Thankachan et al., 2005) (Chrobok et al., 2016)

No accession number is given in the quoted material for the vLGN transcriptomic study, and no IGL-specific patch-seq, single-cell RNA-seq, project-seq, or spatial transcriptomic accession is provided in these references. Sabbagh et al. do explicitly frame their analysis as a transcriptional and spatial classification of vLGN neurons, and they discuss patch-seq as a useful multimodal comparison approach rather than presenting an IGL patch-seq dataset themselves. (Sabbagh et al., 2020) (Gouwens et al., 2020)

Overall, the most evidence-based summary is that IGL neuron heterogeneity is well supported at the level of **two major GABAergic peptide/projection classes**, is further complicated by **light-response and NPY-pattern-defined subclasses in the IGL/VLG region**, and likely sits within a broader **retinorecipient geniculate transcriptomic diversity** that has been demonstrated in vLGN but not yet cleanly resolved for IGL proper in the provided literature. (Chrobok et al., 2016) (Blasiak et al., 2013) (Thankachan et al., 2005) (Sabbagh et al., 2020)

Evidence

(Chrobok et al., 2016)

Two distinct subpopulations of neurons in the thalamic intergeniculate leaflet identified by subthreshold currents.

"The intergeniculate leaflet (IGL) is a flat retinorecipient thalamic structure implicated in orchestrating circadian rhythm, historically considered to be a subdivision of the neighboring ventrolateral geniculate nucleus (VLG). IGL consists of two main neuronal subpopulations: enkephalinergic and neuropeptide Y (NPY)-synthesizing cells. These cell types have different functions, connectivity and firing pattern in vivo, which suggest that they have different membrane currents to support their functional differences"

".Our results suggest that IGL neurons can be divided into two subpopulations based on two ionic currents. A T-type calcium current (IT) was identified in neurons that do not synthesise NPY, whereas all NPY-positive neurons were found to express a marked A-type potassium current (IA)."

(Blasiak et al., 2013)

Differential firing pattern and response to lighting conditions of rat intergeniculate leaflet neurons projecting to suprachiasmatic nucleus or contralateral intergeniculate leaflet.

"The intergeniculate leaflet (IGL) of the lateral geniculate body in the rat is a population of GABAergic neurons that can be divided into two, anatomically and neurochemically distinct populations. One population comprises neuropeptide-Y (NPY)-positive neurons that form the geniculohypothalamic tract innervating the suprachiasmatic nuclei (SCN) and the other population comprises enkephalin-positive (ENK) neurons giving rise to the geniculo-geniculate tract innervating the contralateral IGL (cIGL)"

".All IGL neurons identified as projecting to the contralateral IGL displayed infra-slow oscillatory activity (ISO; i.e. slow rhythmic bursts of action potentials). ISO of these neurons was sustained in the light and was diminished in the darkness. In contrast, all IGL neurons identified as projecting to the SCN displayed a low level of firing in the light and a majority of these cells increased firing in the darkness. All IGL neurons projecting to the SCN were characterised by an irregular pattern of firing in the light and dark. These data are the first to demonstrate that differentially projecting rat intergeniculate leaflet neurons are characterised by distinct firing patterns and opposite responses to light and dark conditions."

(Thankachan et al., 2005)

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"The thalamic intergeniculate leaflet (IGL) is involved in mediating effects of both photic and nonphotic stimuli on mammalian circadian rhythms. IGL neurons containing neuropeptide Y (NPY) have been implicated in mediating nonphotic effects, but little is known about those involved in photic entrainment"

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Diverse GABAergic neurons organize into subtype-specific sublaminae in the ventral lateral geniculate nucleus

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". In this study, we describe subtypes of transcriptomically distinct GABAergic neurons that populate the vLGN and organize into discrete, adjacent sublaminae in the vLGNe. Taken together, our results show four subtype-specific sublaminae of retinorecipient neurons in vLGNe"

". Here, we used spatial distribution and transcriptional profiles to classify neurons into distinct subtypes. Molecular markers remain at present a leading characteristic for such classifications, though they are not without their limitations. It may well be that vLGN neurons can be further subdivided if classified by 2-3 molecular markers rather than one (as we observed in Spp1 + and Pvalb + neurons). However, the very fact that differential expression of one molecule was sufficient to differentiate two vLGN populations is a strong indicator that the nucleus as a whole is more diversely populated than previously appreciated"

". A set of recent studies represent a major step towards this goal by utilizing Patch-seq to simultaneously characterize cortical GABAergic neurons electrophysiologically, morphologically, and transcriptomically (Gouwens et al., 2020)Scala et al., 2020)."

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